

ALISTAIR RODRIGUES

AI Engineer | LLM Applications, RAG and AI Agents

+91 89994 53945 | alistair7@gmail.com | linkedin.com/in/alistair77
github.com/Alistair77 | alistair77.github.io/professional-portfolio-website

SUMMARY

AI engineer with an MSc in Advanced Computer Science with Artificial Intelligence from the University of Strathclyde. I build LLM applications that are meant to be trusted, not just demoed: agents that ask a human before doing anything risky, RAG that says "I don't know" instead of guessing, and eval pipelines that catch regressions before they ship. A background in data analysis and UI/UX design means I care about the whole product, from clean data to an interface people can actually use.

SKILLS

AI and LLMs: LLM application development, AI agents and tool use, RAG, LangGraph, Claude API, Ollama (local models), prompt engineering, LLM evaluation

Backend: Python, FastAPI, REST APIs, PostgreSQL, SQL, Docker, Git, pytest

Frontend and Design: React, JavaScript, HTML/CSS, Three.js, Figma, UI/UX design, usability testing

Data: Pandas, NumPy, Scikit-learn, Matplotlib

Ways of Working: Agile/Scrum, stakeholder communication, team leadership

PROJECTS

Agent Orchestrator | *Python, LangGraph, Claude API, FastAPI, PostgreSQL*

- Built a multi-step AI agent that plans, acts and reviews its own work, where the model can only request actions and a human has to approve anything risky before it runs.
- Logged every decision to an append-only PostgreSQL audit trail, so any run can be replayed step by step weeks later.
- Served it through an async FastAPI API with 19 passing tests and a demo mode that runs the full system without an API key.

ragstar: Hybrid Search RAG | *Python, Ollama, vector search, BM25, reranking*

- Built a fully local RAG system that answers questions about your own documents with clickable citations, combining vector and keyword search and reranking the results for accuracy.
- Added a refusal gate set from measured retrieval scores, so it declines questions the documents can't answer instead of making something up.
- Wrote an evaluation suite that scores retrieval, faithfulness, correctness and refusal against an 18-question golden set.

Agent Evals | *Python, Claude API, PostgreSQL, k-means clustering*

- Built an automated pipeline that checks whether a new version of an AI agent got worse, using the agent's real conversation logs as test data.
- Grouped similar prompts with k-means, used Claude as a judge to write golden answers and rubrics, then scored each new version 1 to 5 on correctness, groundedness and safety.

Jericho: Live AI Agent Dashboard | *Claude Code, Three.js, JavaScript, Web Audio API*

- Built a live 3D dashboard showing which of 30 Claude Code skills in an AI video workflow is running, what it's working on and how far along it is.
- Connected it to real agent activity through a Claude Code hook that records every skill start and finish, so the view only moves when actual work is happening.
- Shipped as a single HTML file with no install or build step, with all sound generated live in the browser.

Semantic Cache | *Python, FastAPI, FAISS, Claude API*

- Built a caching layer in front of Claude API calls that answers repeated or reworded questions from a vector index instead of paying for a new model call.
- Reached a 64% cache hit rate on a 50-prompt benchmark; against the live RAG service, a cached answer came back in 84 ms versus 34 seconds uncached. Covered by 25 tests.

PROFESSIONAL EXPERIENCE

UI/UX Intern, CMS Info Systems Ltd | Navi Mumbai, India | Mar 2026 – Sep 2026

- Designed wireframes and interactive prototypes for internal IT tools, working closely with developers so designs stayed realistic to build.
- Ran usability reviews using heuristic evaluation and flow mapping, and turned the friction points into specific redesign proposals.
- Worked in Agile sprints, joining stand-ups and design reviews with the IT team from kickoff to release.

Events Team Leader, Edinburgh International Conference Centre | Edinburgh, UK | Jan 2024 – Dec 2024

- Led crew teams at high-profile international conferences, handling task allocation and keeping each event on schedule.
- Acted as the main client contact on event days, resolving delegate issues and escalations under pressure.
- Reviewed post-event feedback to find recurring service gaps and fed the fixes into planning for the next event.

Data Analysis Intern, Binex Controls | Pune, India | Dec 2022 – Jul 2023

- Cleaned and prepared datasets in Python (Pandas, NumPy), fixing missing values and inconsistencies before analysis.
- Built Matplotlib charts showing trends across datasets for internal reporting.
- Presented findings monthly to non-technical stakeholders and worked with developers to feed results into internal tools.

EDUCATION

MSc Advanced Computer Science with Artificial Intelligence, University of Strathclyde | Glasgow, UK | Jul 2025

BSc Computer Science, Savitribai Phule Pune University | Pune, India | 2019 – 2022

Relevant coursework: Data Structures, Algorithms, Machine Learning, Statistical Analysis, Database Systems

Secondary Schooling, Saint Ursula's High School | Pune, India

VOLUNTEERING

Data Volunteer, Umbrella Health Partnership (Birmingham City Council Youth Service) | Aug 2023 – Present

- Built the organisation's first data system, turning paper records into a clean, structured dataset leadership could actually use.
- Ran quarterly analysis on it, surfacing trends and gaps that informed programme planning.